



Contents lists available at ScienceDirect

Molecular Aspects of Medicine

journal homepage: www.elsevier.com/locate/mam

Single-cell RNA sequencing: Technical advancements and biological applications

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ARTICLE INFO

Article history:

Received 19 May 2017

Received in revised form

19 July 2017

Accepted 24 July 2017

Available online xxx

ABSTRACT

Cells are the basic building blocks of organisms and each cell is unique. Single-cell RNA sequencing has emerged as an indispensable tool to dissect the cellular heterogeneity and decompose tissues into cell types and/or cell states, which offers enormous potential for *de novo* discovery. Single-cell transcriptomic atlases provide unprecedented resolution to reveal complex cellular events and deepen our understanding of biological systems. In this review, we summarize and compare single-cell RNA sequencing technologies, that were developed since 2009, to facilitate a well-informed choice of method. The applications of these methods in different biological contexts are also discussed. We anticipate an ever-increasing role of single-cell RNA sequencing in biology with further improvement in providing spatial information and coupling to other cellular modalities. In the future, such biological findings will greatly benefit medical research.

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1. Introduction

In 1977 Frederick Sanger invented Sanger sequencing, a DNA sequencing method that utilizes modified dideoxynucleotides to cause chain-termination (for which he was awarded his second Nobel prize in 1980). For the first time, we could read our genetic code. This method is however expensive and labor-intensive. Thus, it took 13 years and cost three billion dollars to complete the “Human Genome Project” in which essentially the entire human genome was sequenced by 2003 (filling in the gaps of the first draft that was published in 2001) (<https://www.genome.gov/10001772/all-about-the-human-genome-project-hgp/>). Sequencing for an individual research project was not economically practical (microarrays were the choice of method) until next generation sequencing (NGS) methods were marketed in 2005. NGS enables high-throughput sequencing by synthesis (SBS), reduces cost and increases feasibility. The recently marketed Nanopore devices, the third generation of sequencers, will further push this limit. With sequencing costs continuing to drop, the first single-cell RNA sequencing method (scRNA-seq) was developed in 2009 (Tang et al., 2009).

Individual cells are the basic building blocks of organisms and each cell is unique. Performing bulk RNA sequencing often masks such uniqueness and fails to reveal latent changes. scRNA-seq has emerged as a revolutionary tool to uncover the uniqueness of each cell and thus study biology at a microscopic resolution and address questions that previously could not be answered. In the past several years, scRNA-seq has been applied to reveal new cell types (Grün et al., 2015; Segerstolpe et al., 2016; Treutlein et al., 2014), investigate the dynamics of developmental processes (Li et al., 2017; Petropoulos et al., 2016; Rizvi et al., 2017; Treutlein et al., 2016), identify gene regulatory mechanisms (Jaitin et al., 2016a; Xue et al., 2013) and reveal random allelic gene expression (G. Chen et al., 2016; Deng et al., 2014; Reinius et al., 2016). In this review, we are summarizing recent technological developments in single-cell transcriptomics as well as their biological applications. For computational analysis of single-cell sequencing data, which is beyond the scope of this review, we can recommend several excellent reviews, including (Bacher and Kendziorski, 2016; Grün and van Oudenaarden, 2015; Stegle et al., 2015).

2. Single-cell RNA sequencing technologies

Since the first scRNA-seq protocol was published, we have witnessed a massive expansion in method development (see Table 1). Such efforts are crucial as each method has distinct

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Table 1
Summary of current single-cell RNA-sequencing methods.

Methods	Captured RNAs	cDNA coverage	Amplification technology	UMI	Strand-specific	Early Multiplexing	Reference
Homopolymer tailing-based PCR							
Tang2009	polyA ⁺ RNAs	Full-length with 3'-biased	PCR after polyA tailing	No	No	No	(Tang et al., 2009)
Quartz-seq	polyA ⁺ RNAs	Full-length with 3'-biased	PCR after polyA tailing	No	No	No	(Sasagawa et al., 2013)
SC3-seq	polyA ⁺ RNAs	Full-length with 3'-biased	PCR after polyA tailing	No	No	No	(Nakamura et al., 2015)
SUPeR-seq	polyA ⁺ and polyA ⁻ RNAs	Full-length	PCR after poly-dA tailing	No	No	No	(X. Fan et al., 2015)
MATQ-seq	polyA ⁺ and polyA ⁻ RNAs	Full-length	PCR after poly-dC tailing	Yes	Yes	Yes	(Sheng et al., 2017)
Switching Mechanism at 5' End of RNA Template (SMART)-based PCR							
STRT-seq	polyA ⁺ RNAs	5' tag (TSS)	Template switching-based PCR	Yes	Yes	Yes	(Islam et al., 2011, 2012; Kivioja et al., 2012)
Smart-seq	polyA ⁺ RNAs	Full-length with weak 3'-biased	Template switching-based PCR	No	No	No	(Ramsköld et al., 2012)
Smart-seq2	polyA ⁺ RNAs	Nearly full-length	Template switching-based PCR	No	No	No	(Picelli et al., 2013, 2014)
Drop-seq	polyA ⁺ RNAs	3' tag (UTR)	Template switching-based PCR	Yes	Yes	Yes	(Macosko et al., 2015)
Patch-seq	polyA ⁺ + RNAs	Nearly full-length	Template switching-based PCR	No	No	No	(Cadwell et al., 2015)
		5' tag (TSS)		Yes	Yes	Yes	(Fuzik et al., 2015)
in vitro transcription-based linear amplification							
CEL-seq	polyA ⁺ RNAs	3' tag (UTR)	<i>in vitro</i> transcription	Yes	Yes	Yes	(Hashimshony et al., 2012)
CEL-seq2	polyA ⁺ RNAs	3' tag (UTR)	<i>in vitro</i> transcription and random primer based PCR	Yes	Yes	Yes	(Hashimshony et al., 2016)
MARS-seq	polyA ⁺ RNAs	3' tag (UTR)	<i>in vitro</i> transcription	Yes	Yes	Yes	(Jaitin et al., 2014)
inDrops	polyA ⁺ RNAs	3' tag (UTR)	<i>in vitro</i> transcription	Yes	Yes	Yes	(Klein et al., 2015)
Designed primers-based PCR							
DP-seq	polyA ⁺ RNAs	Specific gene region	PCR using designed heptamer primers	No	No	No	(Bhargava et al., 2013)
Cyto-seq	polyA ⁺ RNAs	3' tag (UTR)	gene-specific primers based PCR	Yes	No	Yes	(H. C. Fan et al., 2015)
MALBAC-RNA	polyA ⁺ RNAs	Full-length	Quasilinear PCR with 7 random MALBAC primers	No	No	No	(Chapman et al., 2015)

advantages and applicability. In general, scRNA-seq methods are composed of four steps: (1) isolation and lysis of single cells or single nuclei, (2) reverse transcription, (3) cDNA amplification and (4) sequencing library preparation. The first three steps are often finished in one-tube reactions to minimize loss of material. At this time, the last step is commonly conducted using transposase Tn5-based fragmentation for library preparation (if cDNAs are not already in short fragments) followed by sequencing in the Illumina platform. Next, we will elaborate on the first three steps of the scRNA-seq procedure.

2.1. Single-cell isolation and capture

Isolation of single cells is the first and crucial limiting step in scRNA-seq. Cells need to be freed from extracellular matrix and cell-cell adhesion (except for circulating cells), which is often achieved by enzymatic treatment, but can also be accomplished by laser capture microdissection (LCM) and patch clamping (Fig. 1). It is ideal to keep such enzymatic treatment quick and effective so that single cells are dissociated while the RNA remains intact. However, it is important to note that longer time in enzymatic treatment is preferred to physical dissociation due to insufficient enzymatic digestion. Once a single cell suspension is achieved, several methods are used to deliver each cell into individual reaction tubes. Micro-pipetting (manual or automated) is often used to capture cells from samples with few available cells. Although this approach is time consuming and low throughput, visual confirmation under the microscope to ensure that only single cells of high quality are selected is an advantage. Patch-seq, in which cytosol w/

two nucleus can be collected after electrophysiology for gene profiling is also beneficial for samples with few cells of interest (Fig. 1, Table 1). When large numbers of cells are available, fluorescent activated cell sorting (FACS) and microfluidic devices are the methods of choice for cell collection. Both approaches can be high throughput. FACS can also be used to enrich certain cell types based on reporter gene expression, or antibody staining against surface proteins or intracellular markers. However, the occasional inclusion of more than one cell (doublets) and potential damage to cells due to the sorting pressure should be taken into consideration. Microfluidic devices, in which cells are captured in reaction chambers or droplets followed by highly standardized and automated nanoliter reactions are much gentler to cells and also cost-efficient. The drawback is the putative loss of cells and bias for certain cell sizes. Commercially available microfluidic devices include the Fluidigm C1 system, the 10X Genomics Chromium and the recently developed Illumina Biorad SureCell system (with ddSEQ cell isolator). To overcome the cell size selection issue, microwell platforms can be an alternative. By dispensing the cell suspension in a microwell plate there is no bias for size and microscopical visualization can be done to exclude doublets. Color selection is also possible using the commercial platform ICELL8 Single-cell System from WaferGen.

Most cell capture techniques require intact single cells dissociated from fresh tissues. Dissociation of live cells can be problematic when for example mature neurons with extensive cellular processes and high inter-connectivity are to be collected. The needed enzymatic treatment is a cellular stressor, which can degrade surface proteins and may alter gene expression. It also creates a bias

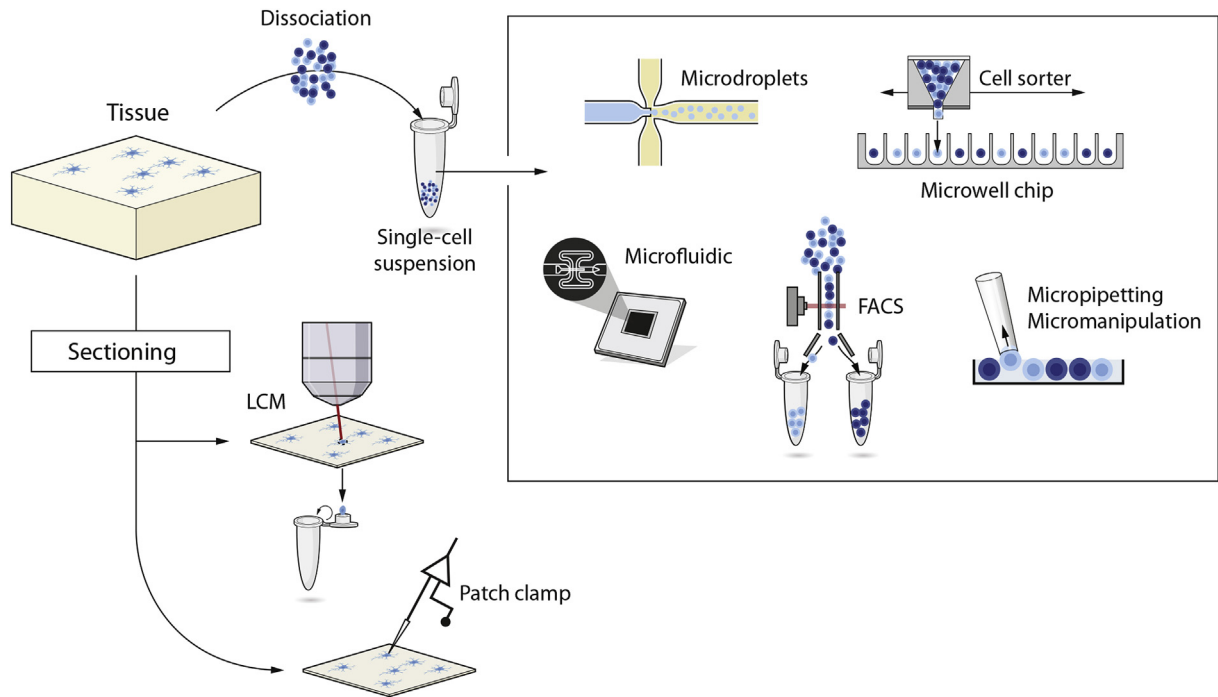


Fig. 1. Illustration of different approaches to isolate and capture single cells for single-cell RNA Sequencing.

towards easily dissociated cell types and cells that are resilient (and don't degenerate in response) to harsh treatments. The applications are widened by the ability to instead isolate cell nuclei only from either fresh or frozen/fixed tissues. Several protocols are available now for this type of procedure (Habib et al., 2016; Krishnaswami et al., 2016; Spalding et al., 2013). For scRNA-seq of nuclei it is important to confirm that cytoplasm has been completely removed, using microscopical visualization prior to single nucleus sorting. Nuclei are sticky and thus challenging to separate. So far, only FACS has been used for this purpose. Compared to whole cells, nuclei contain unprocessed mRNAs and significantly fewer mRNAs than whole cells do (Grindberg et al., 2013).

2.2. Reverse transcription

Capture of RNA and synthesis of cDNA start with reverse transcription. The majority of published protocols use oligodT priming, which selects for polyadenylated mRNAs and some lncRNAs (about 40%) (Yang et al., 2011) (Table 1). This approach avoids capture of ribosomal RNA (rRNA), which accounts for over 95% of the total RNA in mammalian cells (Slomovic, 2006). However, priming through oligodT inevitably precludes non-polyadenylated RNA species, which include most non-coding RNAs and circular RNAs. SUPeR-seq has been developed to sequence both polyadenylated and non-polyadenylated RNAs from individual cells with minimal contamination from rRNAs and genomic DNA (X. Fan et al., 2015). This method uses random primers with fixed anchor sequences (AnchorX-T₁₅N₆). These primers contain both six random nucleotides and oligodT to anneal with polyadenylated and non-polyadenylated sequences. Surprisingly, without any rRNA masking/depletion step, there is no significant amplification of rRNAs and only 1.5% of total sequencing reads map to rRNAs (Rn5s, Rn5.8s, Rn18s, and Rn28s). It was speculated that the limited RNA material, specific lysis and the RT procedure is not strong enough to open secondary structures of rRNAs, therefore leading to low RT efficiency of rRNAs (X. Fan et al., 2015). Unfortunately, this method

has not yet been extensively compared with other popular methods except with the Tang 2009. Very recently, MATQ-seq was reported to be more sensitive and quantitative than Smart-seq2 (Sheng et al., 2017). MATQ-seq provides whole gene body coverage and allows for the detection of total RNA, which is a significant advancement. This improvement is based on the use of MALBAC (multiple annealing and looping-based amplification cycles) primers. These MALBAC primers contain 27 common sequences followed by five random nucleotides which differ only at the 3' end to be either T₂₀ or G₃/A₃/T₃ and greatly increase reverse transcription efficiency and subsequent production of PCR amplicons.

2.3. cDNA amplification

After converting RNA into first strand cDNA, second-strand synthesis can be achieved using different strategies. One approach is the SMART technology (switching mechanism at 5' end of RNA template) that take advantage of transferase and strand-switch activity of M-MLV transcriptase (Moloney Murine Leukemia Virus Reverse Transcriptase) to incorporate template-switching oligos as adaptors for downstream PCR amplification (Zhu et al., 2001). Such methods include Smart-seq, Smart-seq2 and STRT. The other approach ligates the 5' end of cDNA with either poly(A) or poly(C) to build common adaptors for PCR amplification. PCR is a very common approach to generate enough cDNA from minute amounts of input material. However, the exponential amplification can potentially skew the representation of gene expression profiles towards shorter and less G-C rich amplicons.

To avoid this, CEL-seq, which is the first method to use *in vitro* transcription (IVT) as an alternative to acquire linear amplification was developed (Hashimshony et al., 2012). By incorporating the T7 promoter in the polyd(T) primers, IVT is performed after second strand synthesis. Complementary RNAs are either fragmented and ligated with a 3' adaptor (CEL-seq, MARS-seq) or undergo another round of reverse transcription with random primers to improve sensitivity (CEL-seq2). IVT eliminates the requirement for a

template-switch step.

In addition, the MALBAC-RNA method is based on Multiple Annealing and Looping-Based Amplification Cycles (MALBAC) that was developed for whole genome amplification of individual cells (Zong et al., 2012). This method uses quasilinear instead of exponential amplification to amplify and sequence the transcriptomes of single cells. The MALBAC primers generate amplicons to have complementary ends that loop and prevent DNA from being copied exponentially, thereby avoiding amplification bias.

2.4. Which method to choose and how many cells to sequence

More than ten methods have been developed for scRNA-seq (see Table 1). In general, the methods can be categorized into two types; full-length and tag-based. Full-length methods try to achieve a uniform gene body read coverage and increase the number of mappable reads. Such broad coverage potentiates analyses of the data for isoform discovery, splicing events and SNP identification for allelic gene expression. The major drawbacks of full-length methods used to be that it was not possible to multiplex and pool all samples for one tube preparation of Illumina sequenceable libraries, which increased cost and labor. Therefore, these methods were not compatible with high-throughput platforms such as Drop-seq, inDrop-seq or Microwell-seq. More importantly, it was not possible to combine full-length methods with UMIs (unique molecule identifiers) to digitally quantify transcript molecules. UMI is a sequence of several nucleotides which can be integrated into each transcript by reverse transcription (Kivioja et al., 2012). When provided with a large amount of UMIs, each transcript ends up with a random unique barcode, therefore the same random barcode sequence only arises due to PCR amplification. By counting the unique barcodes, the number of copies of a transcript can be accurately evaluated. Interestingly, the recently developed MATQ-seq method can incorporate barcodes and UMIs into MALBAC primers, thus overcoming the afore-mentioned disadvantages.

Most scRNA-seq methods are tag-based and can be further divided into two categories depending on if they utilize a 3' (e.g. CEL-seq/CEL-seq2, MARS-seq) or a 5' tag (e.g. STRT). The main advantage of tag-based methods is that these can be combined with UMIs, enabling multiplexing of more samples and improvement of gene-level quantification and throughput. So far, tag-based methods have relatively low sensitivity as mappable reads are restricted to one end of the transcript. Thus, tag-based methods are mostly used for gene expression quantification and cannot be utilized for isoform identification or splicing.

Therefore, which method to choose depends on the biological questions. For cell-type discovery and characterization of tissue composition, both full-length and tag-based methods are suitable. Tag-based methods allow for pooling of all samples after the RT step, which renders it cheaper and large-scale. For specific purposes such as allelic gene expression analysis and different isoform discovery, full-length methods are the most suitable. Two recent publications have systematically compared several frequently used methods regarding method performance in sensitivity, accuracy and cost efficiency, thus enabling researchers to make well-informed choices (Svensson et al., 2017; Ziegenhain et al., 2017). Not surprisingly, Smart-seq2 outperformed other methods in sensitivity and reproducibility, partly due to increased mappable reads compared to tag-based methods. However, to lower cost, Smart-seq2 relies on in-house Tn5. Drop-seq is preferable when sequencing large number of cells, reaching around 4000 detected genes. More importantly, sensitivity is also a function of sequence depth, which should be taken into account when comparing different methods. Notably, for all methods, when sequencing depth has reached 1M reads/sample there is somewhat of a plateau

in sensitivity, with the increase in read depth from 1M to 4.5M reads per sample only resulting in marginally increased sensitivity (Ziegenhain et al., 2017). However, this is only valid for quantification of gene expression. For splicing discovery, deeper sequencing is warranted and no estimation of the minimal sequencing depth has been done so far.

Another frequently asked question is how many cells are required for analysis. The simple answer is that it depends on how rare the cell types in question are. A rough calculation formula $P(d) = 1 - (1-s)^n$ to estimate the number of cells needed was provided by Nicholas E. Navin, where 'P(d)' is detection power, 's' is equivalent to the subclonal frequency, and 'n' is equal to the number of single cells needed to be sequenced (X. Chen et al., 2016). Based on this calculation, if the interested subtype accounts for 1% of the cells in a region or tissue, 250 cells are needed to reach 0.9 detection power and 500 cells are needed for 1.0 detection power. In addition, replicate experiments are required in order to assess false positive and false negative rates. The number of cells required and the depth of sequencing necessary are also dependent on the distinction of the cell in question compared to the other cells in the population. If a cell has a highly unique transcriptional feature this will be revealed with fewer cells and at a shallower sequencing depth.

2.5. Technical challenges in scRNA-seq

Due to scarcity of RNA material in a single cell and the nature of stochastic transcription, scRNA-seq data often faces challenges of high cell-to-cell variability (i.e. inflated zero expression mainly in lowly expressed genes having 100–1000 read counts). Such variability is caused by both technical noise (e.g. RNA capture efficiency, random dropouts during library preparation) and biological noise (e.g. stochastic gene expression, various cellular states, cell sizes and cell cycle state). Furthermore, experimental batch effects as systematic errors are often found in high-throughput data (Hicks et al., 2015; Leek et al., 2010). Batch effects can be introduced during sample handling steps, i.e. cells are captured and sequenced separately at various sequencing depth, different lots of reagents are used and several biological specimens are involved. Careful planning of each experimental step and the use of multiple biological replicates can minimize batch effects. However, variable gene detection rates across cells due to variable capture efficiency and genetic background of biological specimens are difficult to control by experimental procedure. One way to identify if there are batch effects is to visualize the data using principal component analysis (PCA) and observe if cells are grouped by their experimental origin. Furthermore, multiple batch correction methods have been established to deal with these issues (Johnson et al., 2006; Leek, 2014; Risso et al., 2014; Shaham et al., 2017; Tung et al., 2016). To better account for technical variability due to random dropout events during library preparation, it has become common to add spike-in RNA standards of known abundance to the endogenous samples (e.g. 92 spike-ins developed by the External RNA Control Consortium (ERCC)) (ERCC, 2005a, 2005b). ERCC spike-ins consist of a broad concentration of synthetic RNAs of various lengths and GC content, which are often used to measure biases in RNA-seq experiments and for quality control or normalization (Brennecke et al., 2013; Grün et al., 2014; Kim et al., 2015; Stegle et al., 2015). However, ERCC RNAs have a number of limitations as their molecular properties are different to endogenous RNAs and they also suffer from sampling effects.

Therefore, minimizing the RNA loss and maintaining the fidelity of information is a critical challenge in scRNA-seq technologies. scRNA-seq results should thus be interpreted with caution and functional validation is often recommended.

3. Biological applications

Various biological contexts have been studied using scRNA-seq and the more detailed understanding of cellular heterogeneity revealed is expected to be of considerable biological and clinical relevance. In this review, we focus on findings from early preimplantation development and cerebral cortex where scRNA-seq has led to great conceptual advances (Fig. 2).

3.1. Preimplantation development

Life begins with a single zygote and the differentiation process of the zygote is regulated at the transcriptional level to generate three major cell lineages (i.e. trophoblast (TE), the pluripotent epiblast (Epi) and primitive endoderm (PE)) in the blastocyst. Long-standing questions regarding this process include (1) When do differences arise between individual blastomeres? (2) How are the three cell lineages segregated in time? (3) When is zygotic genome activation initiated? (4) Are there species differences in early

specification events? Several studies have been conducted in mouse and human preimplantation embryos to address these questions.

As blastocysts contain only 60–70 cells in mice and 200–300 cells in humans prior to implantation, scRNA-seq is an excellent tool to decipher gene expression dynamics during preimplantation development at a microscopic resolution. Blastomeres of the early mouse embryo were thought to be equivalent in their developmental properties until the morula stage when they are spatially organized into outer and inner cells. Experimental manipulation of early stage blastomeres, including removing or adding single cells, does not affect embryo development, indicating that early blastomeres undergo a regulative development (i.e. blastomeres have the potential to become any cell type under inductive signals) (Bedzhov et al., 2014; Cockburn and Rossant, 2010). However, using scRNA-seq it was shown that molecular asymmetry within blastomeres could be detected as early as the 4-cell stage (Biase et al., 2014; Xenopoulos et al., 2015). Interestingly, regulated genes were often bi-modally expressed between

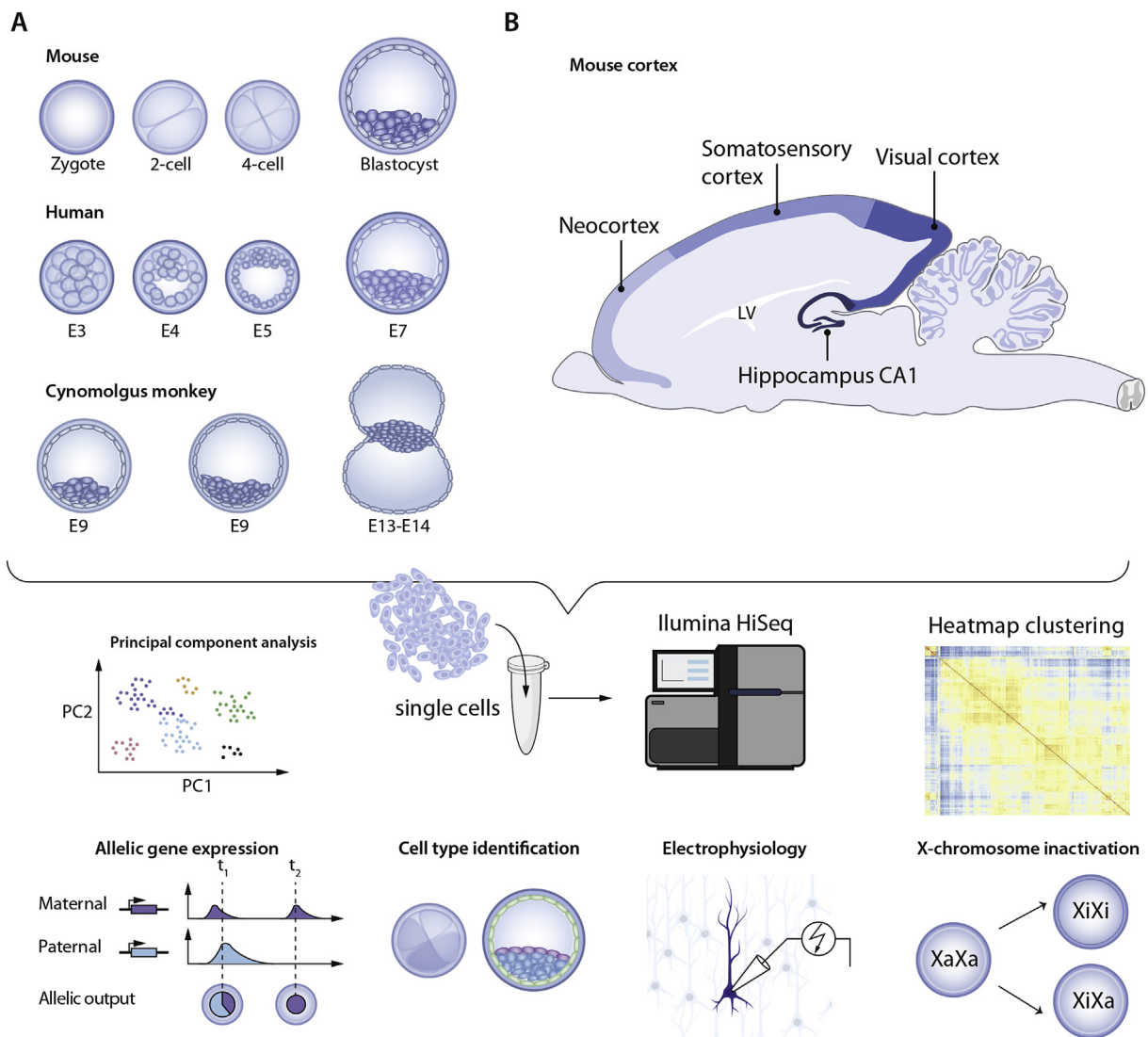


Fig. 2. Summary of selected biological applications using single-cell RNA sequencing technologies. (A) Schematics of preimplantation development in mouse, human and cynomolgus monkey. For cynomolgus monkey, early (embryonic day 13–14; E13–14) postimplantation embryos are also included. (B) Schematics of the mouse brain showing areas of neocortex, somatosensory cortex, visual cortex and the hippocampal CA1 region included in the studies cited. By using single-cell RNA sequencing technologies and computational analyses, novel cell types/states were discovered; allelic gene expression was annotated; and X-chromosome inactivation was revealed.

blastomeres, which is reproducible and due to embryonic genome activation and RNA degradation, but not random noise. Such inter-blastomere differences are related to TE versus ICM fate choice (Biase et al., 2014). Sox21 was later identified to show consistently heterogeneous expression at 4-cell stage. Live tracing of Sox21 progeny confirmed its biased cell differentiation towards inner cell mass (ICM) versus TE (Xenopoulos et al., 2015). Although the precise timing and mechanisms of lineage segregation has been extensively studied for decades, scRNA-seq analysis has now further advanced our understanding by providing a comprehensive view of dynamic gene expression through the whole process and enabling systematical comparisons between mouse and human. The first description of the transcriptome landscape of human early embryos was provided by Yan et al. (Yan et al., 2013). They sequenced 124 individual human cells from oocytes to blastocysts, revealing distinct global expression patterns of each preimplantation stage as well as differential expression of splice variants and long noncoding RNAs. Moreover, it demonstrated that human epiblast cells have dramatically different transcriptomes from hESCs with 1498 differential expressed genes. In addition to identifying stage-specific genes and splicing isoforms, human and mouse preimplantation embryos from oocytes to morula stage were compared (Xue et al., 2013). This cross-species comparison highlighted the evolutionary conservation in genetic programs during early embryogenesis. However, these two pioneering studies in human blastocysts did not touch upon other crucial events such as the precise timing of lineage delineation and zygotic genome activation. Three years later, the most comprehensive single-cell transcriptome analysis of 1529 individual cells from 88 human preimplantation embryos was published (Petropoulos et al., 2016). These data demonstrated that zygotic genome activation occurs later in humans (at 4-cell stage) than in mice (at 2-cell stage) (Deng et al., 2014). Surprisingly the study showed that three cell lineages are concurrently established at embryonic day five in humans, which is different from the “two-step” lineage specification in mouse (Rossant, 2016). More importantly, in contrast to mouse, both X chromosomes in humans undergo progressive inactivation, showing dampened biallelic gene expression to achieve dosage compensation. All these studies imply that there are several biological differences between mouse and human embryonic development. It is challenging to perform detailed gene function studies in human embryos due to the scarcity of material. Therefore, Nakamura et al. performed comprehensive scRNA-seq of 390 cells in the epiblast of pre- (E7–E9) and postimplantation (E13–E14) embryos of cynomolgus monkeys which are closely related to humans (Nakamura et al., 2016). By cross-comparing with mouse epiblasts and hESCs, they revealed that the cynomolgus epiblast (cyEpi) undergo several transitional changes from specification to post-implantation. Specifically, the preimplantation cyEpi retains properties of the pre-gastrulating mouse epiblast, while the late post-implantation cyEpi displays a high similarity with human and monkey ESCs, which are enriched in NOTCH signaling pathways. This indicated a divergence of epiblast development among mammalian species.

Besides its utility in studying comprehensive gene expression dynamics during lineage specification, full-length scRNA-seq is also a powerful approach to determine allele-specific gene expression (ASE) during preimplantation development and X chromosome inactivation along differentiation. Such ASE readout could be achieved with strain-specific SNPs in 82% of genes from outbred embryos derived from C57BL/6 and CAST/Eij mice. By analyzing ASE across 267 cells ranging from oocytes to late-blastocyst embryos, a novel mode of ASE was revealed (Deng et al., 2014). Such ASE was pervasive (i.e. in 12–24% genes) and showed random and dynamic features distinct from previously known imprinting or clonally

stable ASE (Benitez et al., 2017; Reinius et al., 2016; Reinius and Sandberg, 2015). Using mESCs derived from such outbred blastocysts, early embryonic development could be mimicked by *in vitro* differentiation. For the first time, dynamic and asynchronized X chromosome inactivation along the differentiation process in female cells could be revealed (G. Chen et al., 2016). Systematically examining allelic expression of pluripotency genes, it was demonstrated that the manifestation of allelic expression is the functional output of the expression level.

3.2. Mouse cerebral cortex

Systematic classification of all mammalian neurons is a long-standing goal in the field of neuroscience. Understanding cellular composition of the brain offers the opportunity to decipher function and connectivity. scRNA-seq has been widely applied to dissect cellular complexity and diversity in various brain regions, including the cerebral cortex. Analysis of 3005 single cells isolated from mouse somatosensory cortex and CA1 hippocampus region identified nine major clusters by a biclustering algorithm (i.e. BackSPIN) (Zeisel et al., 2015). These clusters were assigned to neuronal, glial and other cell types *post hoc* by marker gene expression. Interestingly, interneurons displayed great heterogeneity with 16 subclasses identified compared to seven subclasses of somatosensory pyramidal neurons, suggesting that interneurons have much broader functions. Such interneuron diversity was also confirmed by a study that used 24 different transgenic Cre lines to enrich for both abundant and rare cortical neurons in the primary visual cortex of adult mice (Tasic et al., 2016). In total, 1679 single cells were FACSed and analyzed and 49 clusters, of which 23 were GABAergic interneurons and 19 were glutamatergic neurons. Intermediate cell states that expressed more than one cell type marker gene, reflecting a phenotypic continuum were also revealed. GABAergic interneurons only account for 20% of all neurons in the visual cortex, but both studies demonstrated that these cells display great heterogeneity in gene expression. It remains to be further investigated if the cell clusters classified by gene expression profiles correlate with distinct functional properties. Cell type classification by transcription profile lays the foundation to interpret cellular function and connectivity. However, both studies employed tailor-designed, rather than unbiased approaches to perform cell-type classification, suggesting that cell type annotation by transcriptomic data is not straightforward.

To directly link anatomical, morphological and functional properties with gene expression, Patch-seq was developed simultaneously by two labs. This technique combines whole-cell electrophysiological patch-clamp recording with scRNA-seq (Fig. 1 and Table 1). One lab combined patch-clamp with Smart-seq2 and focused on the outermost L1 layer of the neocortex and analyzed 58 neocortical cells (Cadwell et al., 2015). This study first used machine learning to classify cellular morphology from firing patterns, which corresponded very well with classification drawn from gene expression profiles. The 58 neocortical cells could be assigned into two subtypes: eNGCs (elongated neurogliaform cells) and SBCs (single bouquet cells) with distinguishable molecular signatures. Importantly, four disease genes implicated in neuropsychiatric illness (i.e. Npas1, Npas3, Dpp6 and Cplx2) were significantly enriched in SBCs. The other study coupled patch-clamp with STRT-seq and focused on cholecystokinin GABAergic interneurons in layers (L)1/2 of the somatosensory cortex (Fuzik et al., 2015). Here, 45 interneurons could be classified into five subtypes and 38 pyramidal neurons into three subtypes according to their electrophysiological properties and morphology. These eight subtypes correlated to individual clusters identified by scRNA-seq data, thus confirming the validity of the Patch-seq method. Gene profiling

from Patch-seq was useful to study ion-channels and receptor genes as well as predicting neurophysiological phenotypes. Compared to live cell scRNA-seq, Patch-seq captured significantly fewer genes and the throughput is relatively low. However, the versatility of different sequencing approaches holds the promise to in-depth characterize the molecular signature, morphology and function of exceptionally complex tissues at a single-cell resolution.

4. Future outlook

4.1. Spatial transcriptomics

Single molecule fluorescence *in situ* hybridization (smFISH) was an early method developed for the quantification of RNAs in tissues at single cell resolution. The methodology utilizes 20 bp long oligonucleotide probes, labeled by a fluorophore which is visualized as spots using fluorescence microscopy (Raj et al., 2008). Initially, the methodology was highly limited in the number of transcripts that could be simultaneously detected. Usually only three to four transcripts could be detected at a time due to the small number of fluorophores available for parallel use. This limitation was overcome by combinatorial labeling where libraries of probes were divided into groups, each coupled to one of three fluorophores which enabled for differentiation of dots that had single colors, pairs of colors or triplet of colors. The number of transcripts that could now be detected increased to $2^n - 1$ (Levsky et al., 2002). The number of simultaneously detectable genes increased significantly further with the use of seven photo-switchable dyes (compared to four fluorophores) and by combining spatial barcodes with super-resolution microscopy. The higher resolution microscopy enables different subparts of the mRNA molecules to be resolved. This allows for different mRNAs to be identified even if they bind the same probe sets, but in a different sequence (Lubeck and Cai, 2012). This method was further improved by barcoding mRNAs by sequential rounds of hybridization, imaging and probe stripping. Each round of hybridization targets each transcript with probes labeled with a single fluorophore. The sample is imaged and probes subsequently removed by DNase1 treatment and followed by another round of the same FISH probes labeled with different fluorophores. Using four fluorophores (F) eight rounds of hybridization (N), seqFISH can cover the entire transcriptome ($F^N = 4^8$) (Lubeck et al., 2014; Shah et al., 2016). One of the great advantages with smFISH is that the hybridization efficiency is very high with >95% of the mRNAs being detected (Levesque et al., 2013; Lubeck and Cai, 2012). In addition, smFISH can resolve splice isoforms, chromosome loci and single-nucleotide polymorphisms (Levesque et al., 2013; Levesque and Raj, 2013). In a similar fashion, FISSEQ (fluorescent *in situ* RNA sequencing) also uses gene-specific probes to read spatial gene expression. But distinctly from smFISH, FISSEQ utilizes rolling circle amplification to augment fluorescent signals (Lee et al., 2014, 2015). As the method does not use an intensity threshold, very faint signals are detected based on their sequence, while noise is eliminated. However, FISSEQ generates fewer reads than RNA-seq and shows an underrepresentation of abundant genes involved in splicing and translation (Lee et al., 2015). In general, all these fluorescent *in situ*-based methods are relatively labor intensive and costly if the entire transcriptome is to be covered.

Spatial transcriptomics at single cell resolution using laser capture microscopy/dissection (LCM) has recently also been developed. LCM allows for careful dissection of individual cells from snap-frozen tissues that have been thinly sectioned. The methodology has an accuracy of well under a micrometer and can even be used at a subcellular level. Furthermore, LCM can be applied to tissues from any embryonic and adult stage and is

particularly suited for tissues that do not easily dissociate. Cells of interest are identified either by a simple histological staining such as Nissl (Cresyl violet) or by a quick antibody staining against a specific protein. It is also feasible to combine LCM with back-labeling to capture cells with a particular axonal target (Saxena et al., 2009). Transgenic animals using nuclear fluorescent labels can with advantage be used for LCM, while cytosolic fluorescent labels tend to be undetectable.

LCM was initially coupled with global transcriptome analysis to study distinct cellular populations using microarrays (exemplified in (Chung, 2005; Hedlund et al., 2010)) and lately also by RNA sequencing (Kadkhodaei et al., 2013; Pembroke et al., 2015). Until recently the number of cells needed for this procedure was rather large, ranging from hundreds to thousands of cells. Recent method developments by our labs have now enabled scRNA-seq of LCM-isolated neurons (LCM-seq) from both mouse and human post mortem tissues (Nichterwitz et al., 2016). With LCM-seq, we simplified procedures and lowered technical noise as well as costs by conducting direct lysis of isolated cells, eliminating the RNA isolation step usually performed following LCM. Another advantage of this method is that the positional information of each cells is preserved and that no tissue dissociation is needed. Thus, LCM-seq allows for simultaneous investigation of cellular heterogeneity and spatial disparity at the single cell level. The significance of preserved spatial information should not be underestimated as position within a tissue is a key determinant of cell identity. Furthermore, as the cells are left largely undisturbed, and connectivity is intact at the time of dissection, the transcriptome is likely to represent the *in vivo* situation to a higher degree than methods requiring dissociation. Another great advantage with LCM-seq is that scarce and partly degraded tissues (tissues with post mortem time of >10 h were successfully used) can be utilized for analysis. However, a limitation to the method so far is that the RNA becomes somewhat fragmented, even at shorter post mortem times, and thus the gene body coverage is lower than for live cells, precluding in depth analysis of RNA splicing (Nichterwitz et al., 2016). Further optimization of LCM staining procedures is likely to overcome this limitation and is currently being thoroughly evaluated.

An elegant methodology named “spatial transcriptomics” in which the transcriptome can be analyzed in intact tissues sections on slides, without the need for cell isolation was recently developed (Stahl et al., 2016). Here, histological sections are placed on slides that contain reverse transcription primers with unique positional barcodes. Approximately 200 million oligonucleotides are placed within each spot with a diameter of 100 μm and a center to center distance of 200 μm . The tissue is permeabilized and reverse transcription reagents are placed on top of the tissue.

The tissue is eventually enzymatically removed leaving the cDNA coupled to the arrayed oligonucleotides on the slide. Encouragingly, the authors showed minimal diffusion of captured cDNAs, similar to tissue capture by LCM (which has no diffusion). The sensitivity was measured to be approximately 7–8% of smFISH, which has near 100% sensitivity. This should be placed in context of scRNAseq on live cells that has been reported to have a 5–40% sensitivity depending on the method used (Stahl et al., 2016). This method has very high resolution (100 μm) and is a highly time efficient method for receiving spatial resolution of entire sections. However, the method cannot easily reveal cellular heterogeneity. Development of this method into a precise single cell methodology would require uniform section thickness of a single cell layer. This is for obvious reasons not possible for any tissue that is heterogeneous in nature. Considering that neighboring cells of varying sizes will be present throughout a tissue, e.g. in the brain different neuron types, astrocytes, microglia, oligodendrocytes are intermingled; and that cells are not coordinated into perfect layers along either a

coronal or sagittal plane, only a selection of cells (in general larger ones) in a section will be captured at a single cell resolution. Thus, the method will capture cells from single as well as multiple layers residing at a specific 2-dimensional coordinate.

4.2. Single-cell multiomics

Sequencing technologies now enable combined profiling of genome, epigenome, transcriptome and/or protein from the same single cell. Thus, a comprehensive understanding of each cell by integrating information from DNA, RNA, protein as well as epigenetic modifications can be achieved. Examples of methods that have already been developed towards this goal include e.g. “DR-seq” and “G&T-seq” for genomic and transcriptomics analyses, “scTrio-seq” for genomic, transcriptomic analyses together with DNA methylome, “scM&T-seq” for transcriptomics and methylome, and “PEA-qPCR” for protein and a panel of genes (Angermueller et al., 2016; Dey et al., 2015; Genshaft et al., 2016; Hu et al., 2016; Macaulay et al., 2015). Studies analyzing genome and transcriptome simultaneously can correlate chromosomal CNVs (copy-number variations), chromosome fusion and SNVs (single nucleotide variations) in regulatory elements with the corresponding gene expression level. Such studies can also reveal clonal structure and cell subtypes within the population, which directly links genotype variation with phenotype outcomes (Dey et al., 2015; Macaulay et al., 2015). On the other hand, incorporation of transcriptomic and methylomic analysis revealed that the degree of DNA methylation of different functional elements in the genome reflects the expression level of genes in single cells (Angermueller et al., 2016; Hou et al., 2016; Hu et al., 2016). Proximity ligation assay (PLA) and proximity extension assay (PEA) based protein analyses in combination with gene expression is discussed in detail in another review in this issue. In the future, integration of total RNAs, small RNAs, chromatin remodeling and chromatin high-order structure into single-cell multiomics will enable detailed characterization of normal cellular functions and disease processes. For further discussion on this topic, several reviews are available (Bock et al., 2016; Macaulay et al., 2017; Prakadan et al., 2017).

Another emerging research front in the field is to combine systematic gene function assays with scRNA-seq analysis (Adamson et al., 2016; Datlinger et al., 2017; Dixit et al., 2016; Jaitin et al., 2016b). For example, Perturb-seq introduce multiple genetic mutations in individual cells through guide RNAs followed by massive parallel scRNA-seq as a readout to decipher gene functions. Another way of perturbing cellular function with a range of factors can be achieved with the Fluidigm Polaris system by taking advantage of microfluidic chambers to isolate, treat and analyze each cell (Ramalingam et al., 2016). These platforms have been exploited to observe the immune cellular responses to environmental signals (Shalek et al., 2013; 2014; Wills et al., 2017).

4.3. Human cell atlas and precision medicine

In 2016, a group of world-leading scientists started building a collection of cell maps that will cover different human tissues and provide a cellular understanding in health and disease. This collective effort is termed “Human Cell Atlas”, <https://www.humancellatlas.org/>, and so far, includes the immune system, the central nervous system, epithelial tissue, fetal cells and cancer. This extensive atlas will offer a comprehensive reference map of cell types, marker genes, signaling pathways and regulatory mechanisms to the community. An enormous amount of data is expected to be generated and this consortium will also serve as a model for best practices in research coordination and data archiving for further analysis. The U.S. National Research Council (NRC) brought

up the concept of precision medicine in 2011 with the definition “an emerging approach for disease treatment and prevention that takes into account individual variability in genes, environment, and lifestyle for each person”. The extensive Human Cell Atlas, conducted on tissues across individuals and diseases will be exceptionally beneficial for the identification of biomarkers and drug target and to further develop precision medicine.

4.4. To relate transcriptional differences to cell types and functionality

scRNA-seq data has already revealed a cellular heterogeneity in several brain regions and tissues that is greater than previously expected. What lies ahead is now the arduous task of functionally evaluating to what extent this heterogeneity in RNA content between related cells reflects distinct functionalities. Most scRNA-seq studies so far (except for Patch-seq) have been rather descriptive. Thus, it is currently unclear to what extent identified transcriptomic differences reflect distinctions in cellular function and thus actual cell types rather than alternate states of the same cell type. So when does a statistically significant increase in one or more particular mRNA transcript(s) reflect a distinct cellular function? This clearly relies on the function of the gene(s) in question and what other transcripts are present. It also relies on the stability and thus half-life of that particular transcript in each cell and if the rate of translation and specifics of posttranslational modifications. Thus, it is not a simple task to relate function to transcriptional differences without functional testing. However, a combination of e.g. cell and molecular biology, biochemistry, physiology, mathematical modeling is sure to give us the answers to the questions that the revolutionary scRNAseq technologies are unraveling. Thus, we envisage that single-cell technologies will be indispensable tools for providing initial important biological targets for further functional annotation.

Acknowledgements

The work in our laboratories are supported by EU Joint Programme for Neurodegenerative Disease (JPND) (529-2014-7500); Swedish Medical Research Council (2016-02112); Åhlén-stiftelsen (MA1/h16); Birgit Backmark's Donation to ALS research at Karolinska Institutet in memory of Nils and Hans Backmark; Parkinsonsfonden (910/16); Ulla-Carin Lindquists stiftelse för ALS forskning; NEURO Sweden, Magnus Bergvalls stiftelse (2016-01531) and the Strategic Research Area Neuroscience (StratNeuro) to EH; and by Swedish Research Council (E0287001); Svenska Sällskapet för Medicinska Forskning (SSMF); Jeansson's Stiftelser; Åke Wiberg Stiftelser (M16-0015); Karolinska Institutet Foundation (2016fobi47682) to QD. The authors thank Mattias Karlen for his excellent work generating the schematics in Figs. 1 and 2.

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